KEYLOOP TECHNICAL CHALLENGE · ENGINEERING LEAD

Unified Service Scheduler

Scenario A: booking a bay and a qualified technician for the whole visit

Go PostgreSQL Redis Kafka WebSocket React Prometheus

2 What the brief asks for

1

Book

A vehicle, a service, and a dealership, at the time the customer asks for.

2

Check the whole visit

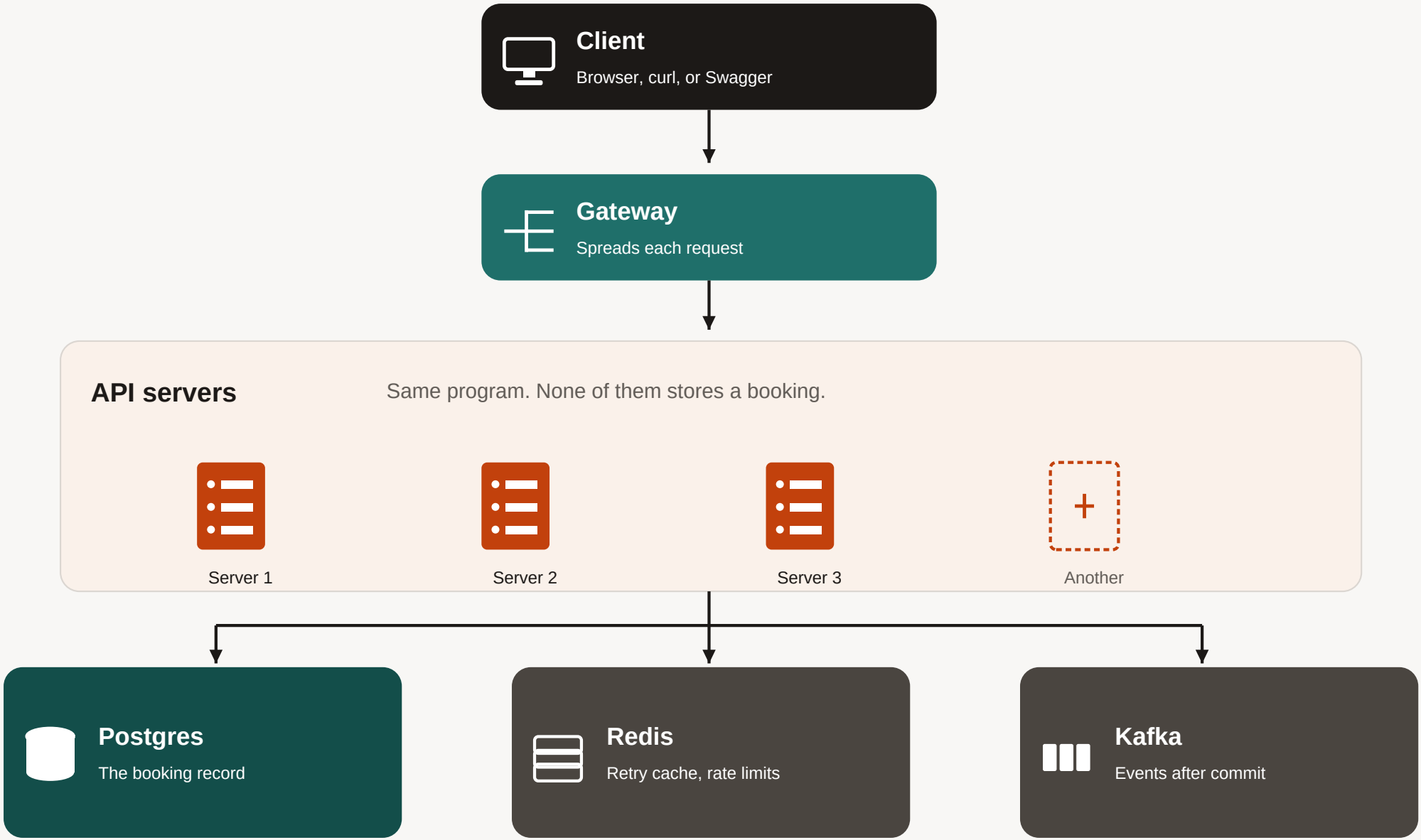
A bay and a qualified technician must both be free from the start until the end.

3

Save the appointment

One record ties together the customer, the vehicle, the technician, and the bay.

3 One gateway, identical stateless API servers



Prometheus scrapes every server; Grafana and alerts on top. Logs are JSON with a request ID and the instance.

4 The event is saved with the booking

1 Commit

Appointment and outbox row, one transaction.

2 Kafka

A worker publishes the row after the commit.

3 Every server

Each server updates browsers with the day board open.

The day board is not the customer message

Confirmation and a change of time go out as email. A reminder 24 hours before, and 2 hours before, goes out as SMS. Locally that is a log line, or a webhook.

5 One booking, in order

1 Ask



Dealership, vehicle, one to five jobs, and a start on the 30-minute grid.

2 Day fits

Open that weekday, clear of lunch, long enough, check-in desk not full.

3 Bay



Express General Diagnostic Alignment

4 Technician



Lubrication Brakes General

5 One move

If nothing is free, move one confirmed job that has not started. Same time.

6 Save

One vehicle, one bay, one technician, for the whole window.

Postgres rejects the second hold

While a visit is confirmed, checked in, or in progress, the same bay, the same technician, and the same vehicle cannot overlap. A lost race is refused (409 or 422). It does not create two appointments.

6 The guarantee is a database constraint

migrations/001_schema.sql

```
CREATE TABLE appointments (
  ...
  CONSTRAINT appointments_bay_no_overlap
  EXCLUDE USING gist (
    service_bay_id WITH =,
    tstzrange(start_at, end_at, '[)') WITH &&
  ) WHERE (status IN (1, 2, 3)),
  -- the same for technician_id and vehicle_id
```

1 Whole duration

Two time ranges on the same bay cannot overlap at any point, not just at the start.

2 Only while holding

1, 2, 3 = confirmed, checked in, in progress. Done, cancelled and no-show free the slot.

3 Race on two servers

The engine checks first. A request that still collides is refused (409 or 422).

Correct even if the application is wrong

Application code can have a bug, and two API servers can race. Postgres still refuses the second hold. TestSlotBurst fires many clients at one slot across two instances: exactly one booking, every loser a clean refusal.

7 How I directed the AI

1 Rules before code

- The double-booking rule lives in Postgres, not in a Redis lock.
- Events go through an outbox, not a direct Kafka write.
- The AI proposed both alternatives. I rejected both, with reasons in the design doc.

2 Guardrails, not prompts

- AGENTS.md: the brief every agent reads.
- Hooks: gofmt, keep the two OpenAPI copies equal, ask before an old migration is edited (CI rejects it).
- A reviewer sub-agent checks booking changes against the invariants.
- CI runs the same checks on every change.

3 An independent check

- AI-written code passes AI-written tests. That proves little.
- The stress run hammers the cluster, then a separate SQL oracle judges every requirement over every row.
- Load testing (stress and bench) found two real bugs. Next page.

Run like a team

A new engineer reads the same file and hits the same checks. The rules do not depend on who, or what, wrote the change. The design choices are mine; the checks decide what is kept.

8 What load testing found, and unit tests missed

Deadlock on a full day

Overlapping bookings of mixed length deadlocked inside Postgres.

p99 latency

6.1 s → 105 ms

Fix

One lock per dealership per day, taken before the insert.

Pool drained on a hot slot

Losing requests held that lock while trying stale candidates, so every other request waited for a connection.

connection pool for every other request

drained → free

Fix

Re-read candidates under the lock; queue in the server before taking a connection. The hot day now waits in memory.

Both are regression tests now

What I learned: with AI, writing code is cheap. Verification is the real work, and it has to come from outside the code being checked.

9 What I would build next

1 Year of the car

Job length follows the VIN year, and how fast that technician works. Today every oil change is 30 minutes.

2 Several technicians

One express job can use more than one person. Today it is still one.

3 Leave the car

Labour hours for that day, placed later. Not one bay held for the whole visit.

4 After the invoice

The repair order closes. Reminders go through a real email and SMS provider.

Phase 2, written in the system design, not in this demo. The brief still needs one bay and one technician free for the whole visit.

10 In one page

1 Stateless servers scale out

No booking in memory. Add a server behind the gateway; nothing splits.

2 The guarantee lives in the database

GiST exclusion on bay, technician and vehicle. A lost race is refused, never doubled.

3 Events cannot be lost

Outbox in the booking transaction, then Kafka, then every open day board.

4 AI output kept only on proof

Rules first, guardrails, and an independent SQL oracle as the bar.

Try it

Run `./scripts/up.sh`

Test `./scripts/verify.sh test`

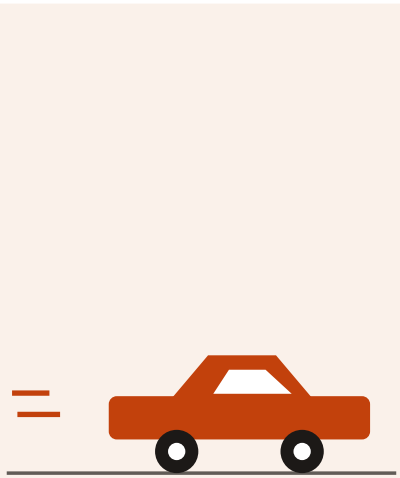
Read `README.md` · `docs/system-design.md` · `docs/slides.pdf` · `/api/v1/docs`

Appendix

Domain rules behind the engine, for questions.

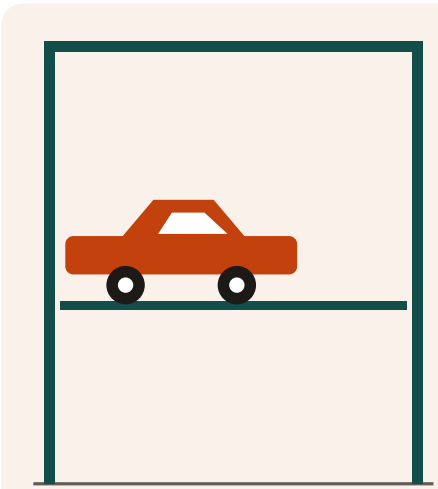
- Bay types and fallback
- Technician skills and how one is chosen
- The three seed jobs
- Moving one job to make room
- What Postgres holds
- Lifecycle, cancel, no-show, late arrival

12 Four bay types



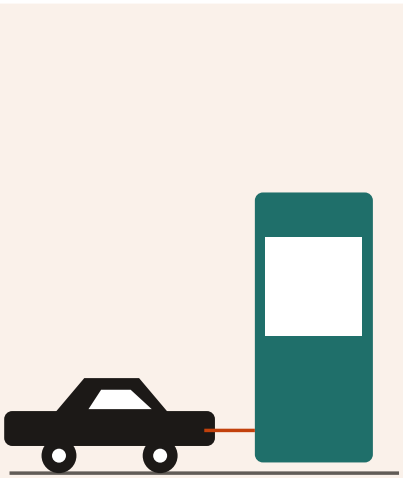
Express

Quick lane for short jobs.



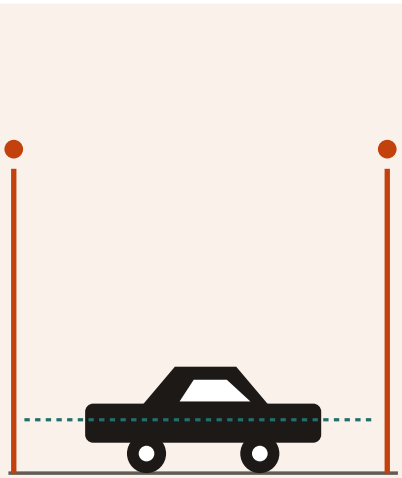
General

Car goes up on the lift.



Diagnostic

A scanner beside the car.



Alignment

Sensors and a beam.

Own type first. Oil may use a general bay if express is full. A short waiter job may use express if general is full. Diagnostic and alignment are never swapped.

13 Own bay type first, then a fallback

Oil change

Express is full

It may take a general bay.



Short waiter job

General is full

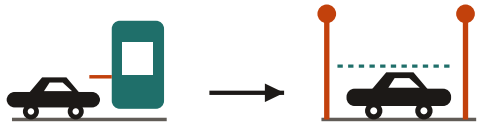
It may take express.



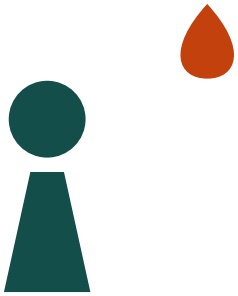
Diagnostic and alignment

Both are needed

Never swapped. Book two visits.



14 Three technician skills



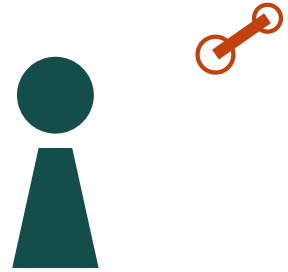
Lubrication

Oil changes.



Brakes

Brake work.



General

The other repairs.

Must have every skill, enough level, on shift. Then lowest level, fewest extra skills, best fit. A brake job does not go to someone who only has the oil skill.

15 How one technician is chosen

1. **Every skill**
   They can do all of the job lines.

2. **Level**
   High enough for the hardest line. Lowest such level is preferred.

3. **On shift**
   Working that day, and not on a break or away.

4. **Best fit**
   Fewest extra skills, then the choice that keeps long free stretches whole.

16 Three jobs in the seed

Oil change

30 min

from level 1

Bay
Express

Skill
Lubrication

Brake service

90 min

from level 2

Bay
General

Skill
Brakes

Major service

180 min

from level 3

Bay
General

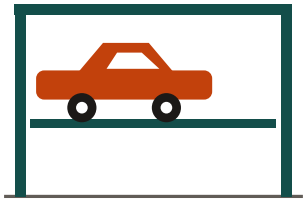
Skill
General

17 If nothing is free, move one job

Before

Bay A is taken

by a confirmed job, not started

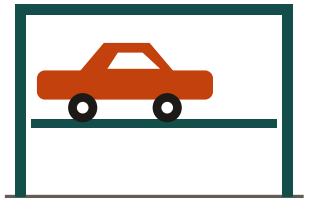


Bay B is free

After, same time

The old job moves to bay B

The new job takes bay A

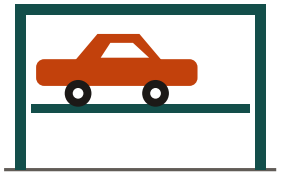


Only a confirmed job that has not started can move. A checked-in job stays where it is. The customer of the moved job keeps the same start time.

18 What is held, and what Postgres rejects



One vehicle



One bay



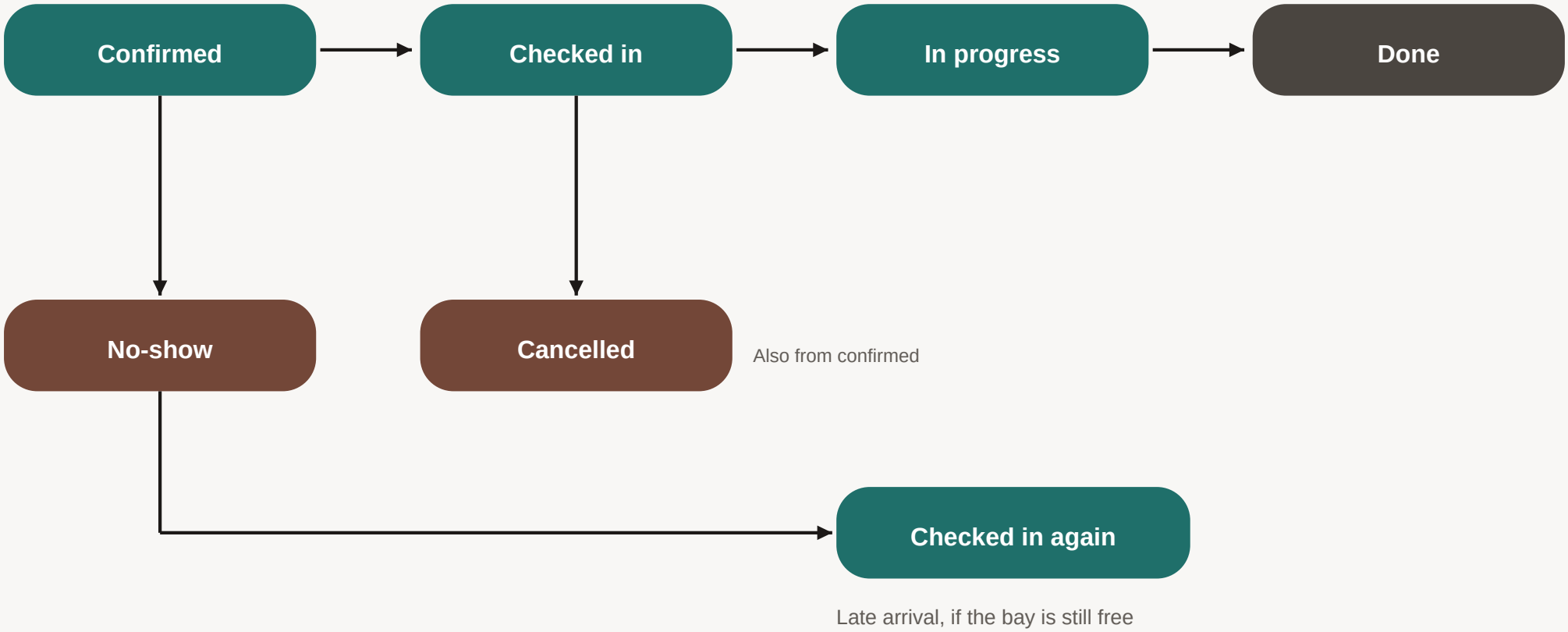
One technician

A second hold on the same one is rejected

While the visit is confirmed, checked in, or in progress, that bay, that technician, and that vehicle cannot overlap another hold. The loser of the race is refused (409 or 422), never given a second appointment.

19 What happens after the booking

Teal keeps the bay, the technician, and the vehicle. Grey and brown release them.



Done, cancelled, and no-show release the bay. Moving the time keeps the same appointment. Changing the bay or the technician does not change the time.

20 Cancel, no-show, and a late arrival

Cancelled

From confirmed, and also after check-in. The bay, the technician, and the car are released.

No-show

Only from confirmed, after the start time, if the customer has not arrived.

Checked in again

A late arrival before the booked end, if that bay and that technician are still free.